Republic of the Philippines

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LAB REPORT I

(ARRAY APPLICATION [2D])

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MACHINE PROBLEM DESCRIPTION:

Finding a Defective Machine in a Factory

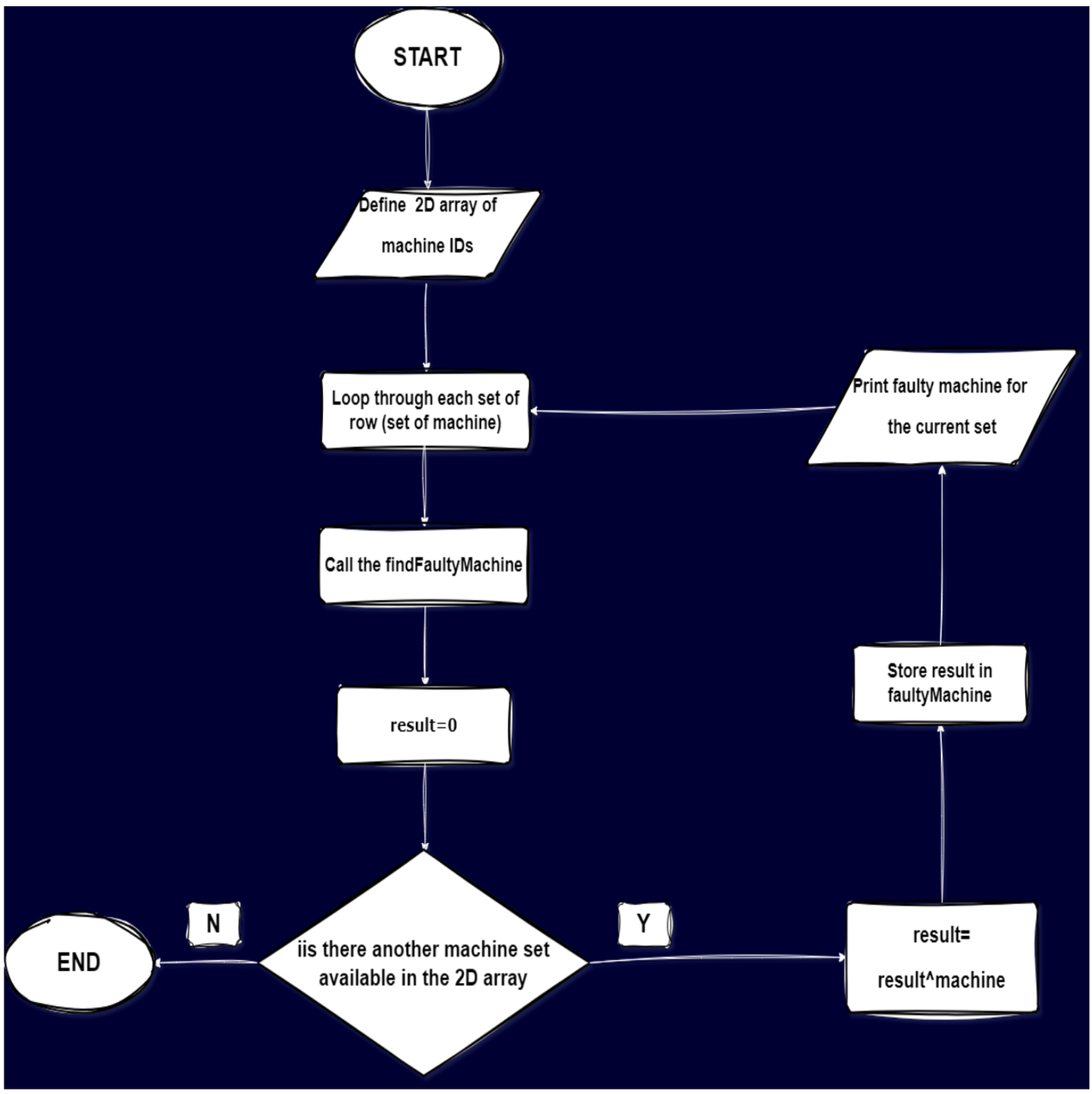
This Python program is a simulation of a factory where a number of machines will be producing products. Each machine has an ID, and as each one is supposed to have appeared twice in the dataset, because it produces an even number of items, but a machine has gone **faulty**, which, on producing items, will result in an odd number of items being produced, so its ID will appear just once in the list. The overall aim of the program is to identify the faulty machine in both the 1D array and the 2D array.

2D Array:

The software will process multiple sets of machine IDs where, in the 2D array version, each set is represented by a row. The software moves on to each set and performs the XOR operation to figure out the faulty machine in each set.

2D Array Flowchart

This flowchart shows how the program processes multiple sets of machine IDs (2D array). Each set is processed individually, and the faulty machine in each set is found using XOR.



SOURCE CODE:

2D Array Code

The code processes multiple sets of machine IDs (2D array) and finds the faulty machine in each set:

findTheFaultyMachine_2D.py

```python
def findFaultyMachine(machines):
    result = 0
    for machine in machines:
        result ^= machine
    return result
def main():
    machines = [
        [101, 202, 303, 404, 101, 303, 404],
        [110, 220, 330, 110, 330],
        [501, 601, 501, 701, 601]
    ]
    for i, row in enumerate(machines):

        faultyMachine = findFaultyMachine(row)
        print(f"Faulty machine in set {i+1}: {faultyMachine}")
if __name__ == "__main__":
    main()
```

2D Array Sample Output:

The three different sets are the machine IDs in the 2D array. The program goes through each set and identifies which of them is faulty.

```
findTheFaultyMachine_2D

"C:\Users\Gian Carlo Trilles\PycharmProjects\pytho
Faulty machine in set 1: 202
Faulty machine in set 2: 220
Faulty machine in set 3: 701

Process finished with exit code 0
```